



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Circles, Arcs & Sectors

Contents

- * Area & Circumference of Circles
- * Arc Lengths & Sector Areas



Your notes

Area & Circumference of Circles

Area & Circumference

What are the properties of a circle?

- A **circle** is a shape that is made up of all the points on a 2D plane that are **equidistant** from a single point
 - Equidistant means the same distance
- The **circumference** of a circle is its **perimeter**
- The **diameter**, d , of a circle is **twice** its **radius**, r
- π (pi) is the number (3.14159 ...) that is the ratio between a circle's **diameter** and its **circumference**
 - You may be asked to give an answer to a question as an 'exact value' or 'in terms of π '

What circle formulae do I need to know?

- The **formulae** for the **area** and **circumference** of a circle are **given to you in your exam**
 - However, it can be useful to remember them!
- The **area of a circle** can be calculated using the formula:
 - $A = \pi r^2$
- The **circumference of a circle** can be calculated using either of the following two formulae:
 - $C = \pi d$
 - $C = 2\pi r$

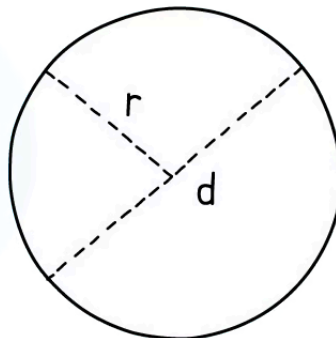


Your notes

CIRCLE

AREA, $A = \pi r^2$

CIRCUMFERENCE, $C = \pi d = 2\pi r$



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- Working with circle formulae is just like working with any other formula:
 - Write down** what you know (or what you want to know)
 - Pick the correct **formula**
 - Substitute** the values in and **solve**

How do I find the circumference of a circle?

- Identify the **diameter**
 - This is **double** the length of the **radius**
- Multiply** the **diameter** by π

How do I find the area of a circle?

- Identify the **radius**
 - This is **half** the length of the **diameter**
- Square** the **radius**
- Multiply** the **radius squared** by π



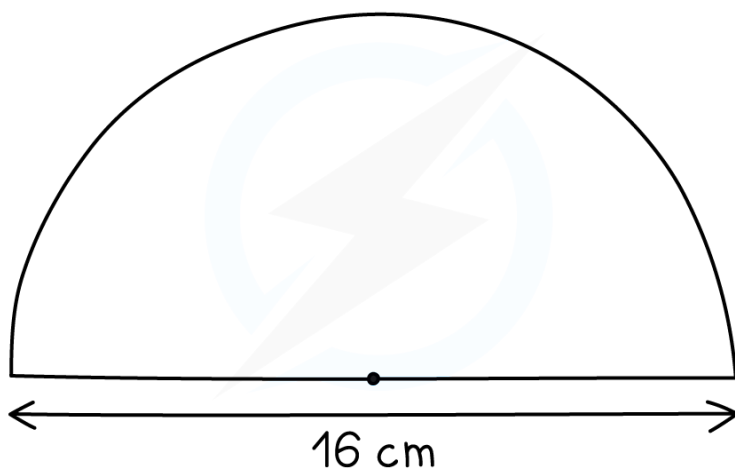
Examiner Tips and Tricks



Worked Example

Find the area and perimeter of the semicircle shown in the diagram.

Give your answers in terms of π .



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The **area** of a semicircle is half the area of the full circle with the same diameter, so begin by finding the area of the full circle

Find the radius by dividing the diameter by 2

$$r = \frac{16}{2} = 8 \text{ cm}$$

Substitute this into the formula for the area of a circle $A = \pi r^2$.

Leave your answer in terms of π , (this just means do not multiply by π on your calculator)

$$A_{\text{full circle}} = \pi(8)^2 = 64\pi$$

Find the area of the semicircle by dividing the full area by 2



Your notes

$$A_{\text{semicircle}} = \frac{1}{2}(64\pi) = \frac{64}{2}\pi = 32\pi$$

$$\text{Area} = 32\pi \text{ cm}^2$$

The **perimeter** of the semicircle is made up of both the arc of the circle (half of the circumference) and the diameter of the semicircle

Find the full length of the circumference of the circle using the formula

$$C = 2\pi r \text{ (or } C = \pi d)$$

Substitute the radius = 8 cm into the formula

Again, leave your answer in terms of π

$$C = 2\pi(8) = 16\pi$$

Find the length of the arc (the curved part of the perimeter of the semicircle) by dividing the full area by 2

$$\text{Curved length} = \frac{1}{2}(16\pi) = \frac{16}{2}\pi = 8\pi$$

Find the full perimeter by adding this to the length of the diameter of the circle

$$\text{Perimeter} = 8\pi + 16 \text{ cm}$$



Your notes

Arc Lengths & Sector Areas

Arc Lengths & Sector Areas

What is an arc?

- An **arc** is a **part** of the **circumference** of a circle
- Two points on a circumference of a circle will create **two arcs**
 - The **smaller** arc is known as the **minor arc**
 - The **bigger** arc is known as the **major arc**

What is a sector?

- A **sector** is the part of a circle enclosed by two **radii** (radiuses) and an arc
 - A sector looks like a slice of a circular pizza
 - The curved edge of a sector is the arc
- Two radii in a circle will create **two sectors**
 - The **smaller** sector is known as the **minor sector**
 - The **bigger** sector is known as the **major sector**

What formulae do I need to know?

- You need to be able to calculate the **length of an arc** and the **area of a sector**
- The **angle** formed in a sector by the two radii is often labelled θ (the Greek letter "theta")
- You can calculate the **area of a sector** or the **length of an arc** by adapting the formulae for the area or circumference of a circle
 - A full circle is equal to 360° so the fraction will be the angle, θ° , out of 360°

- Area of a sector = $\frac{\theta}{360} \times \pi r^2$

- Arc length = $\frac{\theta}{360} \times 2\pi r$



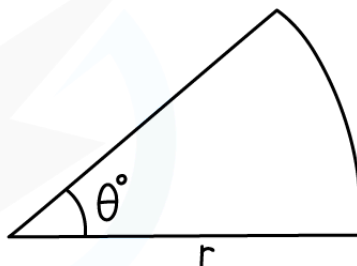
Your notes

SECTOR AREA

$$\text{AREA} = \frac{\theta}{360} \times \pi r^2$$

ARC LENGTH

$$\text{ARC} = \frac{\theta}{360} \times 2\pi r$$



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- Working with **sector and arc formulae** is just like working with any other formula:
 - **Write down** what you know (or what you want to know)
 - Pick the correct **formula**
 - **Substitute** the values in and **solve**

How do I find the length of an arc?

▪ STEP 1

Divide the **angle** by **360** to form a fraction

$$\frac{\theta}{360}$$

▪ STEP 2

Calculate the **circumference** of the **full circle**

$$2\pi r$$

▪ STEP 3

Multiply the **fraction** by the **circumference**

$$\frac{\theta}{360} \times 2\pi r$$



Your notes

How do I find the area of a sector?

- **STEP 1**

Divide the **angle** by **360** to form a fraction

- $\frac{\theta}{360}$

- **STEP 2**

Calculate the **area** of the **full circle**

- πr^2

- **STEP 3**

Multiply the **fraction** by the **area**

- $\frac{\theta}{360} \times \pi r^2$



Examiner Tips and Tricks

- The area and circumference of a circle formulae are given to you in the exam.
 - You just need to remember how to find the correct fraction of the whole circle.

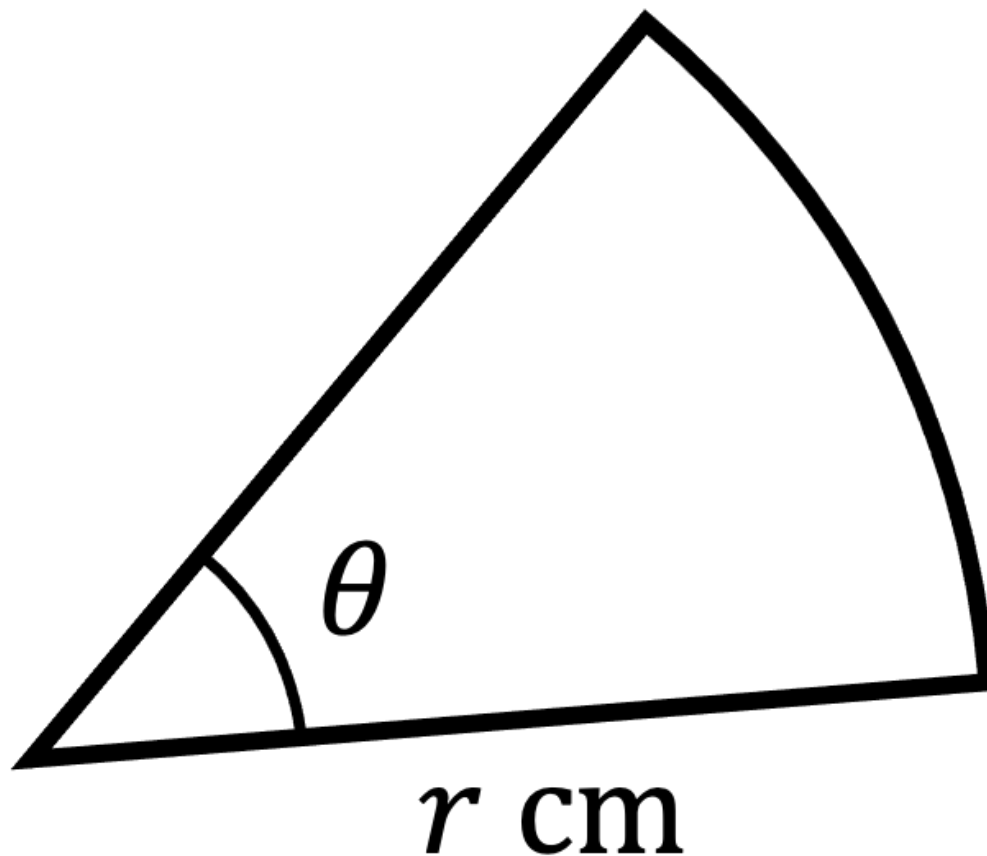


Worked Example

A sector of a circle is shown.



Your notes



The angle, θ , is 72° and the radius, r , is 5 cm.

(a) Find the area of the sector, giving your answer correct to 3 significant figures.

Substitute $\theta = 72^\circ$ and $r = 5$ into the formula for the area of a sector, $A = \frac{\theta}{360} \pi r^2$

$$A = \frac{72}{360} \pi \times 5^2$$

Use a calculator to work out this value

15.70796...

Round your answer to 3 significant figures

15.7 cm²

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(b) Find the length of the arc of the sector, giving your answer as a multiple of π .

Substitute $\theta = 72^\circ$ and $r = 5$ into the formula for the length of an arc, $l = \frac{\theta}{360} 2\pi r$

$$l = \frac{72}{360} \times 2 \times \pi \times 5$$

Simplify the number part without π

$$\frac{72}{360} \times 2 \times 5 = \frac{1}{5} \times 10 = 2$$

Write down the final answer with π

2 π cm